

Srikrishnaa Vadivel

Embedded Systems Engineer | FPGA | Robotics | Sensor Logging

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Summary

Embedded systems engineer with field-tested sensor logging, MCU firmware, FPGA logic, robotics, and scientific instrumentation experience. Comfortable from oscilloscope validation and low-level C through host-side visualization and data systems.

Experience

Probe Technologies / Weatherford, Software and Embedded Engineer 2019–2021

- Programmed C8051-class microcontrollers and Lattice FPGA logic for sensor logging in geological well tools.
- Validated embedded systems with oscilloscopes, bench instrumentation, and field deployments.
- Built host-side CUDA/OpenGL software for large acoustic waveform processing and visualization.

Yale University, Embedded Systems Teaching Assistant / PhD Researcher 2021–present

- Served as embedded systems TA while pursuing PhD research in computational materials and scientific computing.
- Used Linux, C/C++, Python, MPI/OpenMP, PETSc/SLEPc, and HPC systems for simulation workflows.

Cornell ECE Robotics Hand Project 2017

- Built an embedded robotic hand project later published in Circuit Cellar.

Portfolio Projects

- **MIPS Processor:** Verilog single-cycle CPU and self-checking Icarus Verilog test bench.
- **Matrix-Multiply Accelerator:** small AI accelerator block with start/done handshake and test bench.
- **ROS 2 TurtleBot Simulation:** robotics simulation plan with trajectory analysis artifacts.

Education

Yale University, PhD Electrical Engineering, 2021–present

Yale University, MPhil Electrical Engineering, 2023

Yale University, MS Electrical Engineering, 2023

Cornell University, MEng ECE, 2017–2018

Cornell University, BA ECE, 2013–2017

Skills

C, C++, Verilog/SystemVerilog, FPGA, MCU firmware, FreeRTOS concepts, SPI, I2C, UART, ADC logging, oscilloscopes, Linux, Python, CUDA, OpenGL, Git.